

Statistics of Financial Markets

Seminar 16: Review



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AUC**: Area Under the (ROC) Curve
- **BET**: Bucharest Exchange Trading (index)
- **EAD**: Exposure At Default
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **EWMA**: Exponentially Weighted Moving Average
- **FHS**: Filtered Historical Simulation
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **HS**: Historical Simulation
- **JB**: Jarque–Bera (test)
- **LGD**: Loss Given Default
- **MDD**: Maximum DrawDown
- **PD**: Probability of Default
- **ROC**: Receiver Operating Characteristic (curve)
- **SE**: Standard Error
- **SR**: Sharpe Ratio
- **VR**: Variance Ratio

Today's Question and Seminar Structure

- Question of the seminar
  - **can you solve, on paper and on data, a problem from any chapter of the course?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the Exercise Map (next slide)
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Topic	Type	Chapters
A1	returns, drawdown, annualisation, Sharpe ratio	Solved	1
A2	Jarque–Bera, Student-t, Hill, stable scaling	Proposed	2, 3, 5
A3	Ljung–Box, variance ratio, robust test	Solved	7
A4	GARCH, EWMA and Parkinson volatility	Proposed	8, 9
A5	VaR 1%, ES 2.5%, Kupiec, traffic light	Proposed	10
A6	logit PD, expected loss, AUC, ΔCoVaR	Proposed	12, 15
B1	a full check-up of the BET since 2015	Solved	1–11
B2	VaR 1% backtests: S&P 500 and Bitcoin	Proposed	8, 10, 14
B3	efficiency and memory in two windows	Proposed	7, 11
C2	audit an AI answer	Proposed	0–15

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model named in its statement
- Part A: calculations on paper; Part B: real data, in the notebook, each ending with an interpretation question; Part C: the critique of an AI answer

Data Used

Series	Source	Price used	Period
BET, S&P 500	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$, each series on its own calendar; the last day is 18 September 2026
- Part A needs only a calculator; Part B uses the notebook, with the functions of the course already written (`returns`, `moments`, `ljung_box`, `variance_ratio`, `garch_t`, `kupiec`, `hurst_rs`)

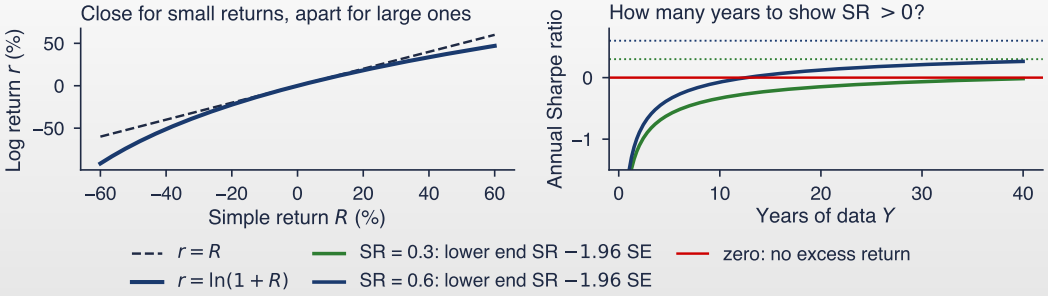
Prerequisites for Today (1/16): Returns, Distributions, Tails

Quantity	Formula
Returns	$R_t = P_t/P_{t-\mathbf{1}} - 1$, $r_t = \ln(1+R_t)$; total: $e^{\sum r_t} - 1$
Annualisation	mean $\times A$, volatility $\times\sqrt{A}$; $A = 252$ (shares), 365 (crypto)
Sharpe, drawdown	$(\mu - r_f)/\sigma$, SE $\sqrt{(1+\mathrm{SR}^{\mathbf{2}}/2)/Y}$; $D_t = P_t/\max_{s\le t} P_s - 1$, recovery $d/(1-d)$
Jarque–Bera	$\frac{n}{\mathbf{6}}S^{\mathbf{2}} + \frac{n}{\mathbf{24}}K^{\mathbf{2}} \sim \chi^{\mathbf{2}}(2)$, $\chi^{\mathbf{2}}_{\mathbf{0.95}}(2) = 5.99$
Student-$t(\nu)$	excess kurtosis $6/(\nu-4)$; unit-variance quantile $t_\nu^{-\mathbf{1}}(\alpha)\sqrt{(\nu-2)/\nu}$
Hill	$\hat\alpha = [\frac{\mathbf{1}}{k}\sum_{i=\mathbf{1}}^{k}\ln(L_{(i)}/L_{(k+\mathbf{1})})]^{-\mathbf{1}}$, SE $\hat\alpha/\sqrt{k}$; moments exist for $p < \alpha$
Stable law	sum of h copies: scale $\times h^{\mathbf{1}/\alpha}$ (Normal: $\sqrt{h}$)

Prerequisites for Today (2/16): Notation for Returns, Distributions and Tails

- Returns and performance
  - P_t: the price at the end of day t; R_t: the simple return; r_t: the log return; A: the number of observations per year
  - μ, σ: the annual mean and volatility; r_f: the risk-free rate; SR: the Sharpe ratio; SE: its standard error; Y: the number of years
  - D_t: the drawdown; $\max_{s \leq t} P_s$: the highest price up to day t; d: the size of a loss, as a fraction
- Distributions and tails
  - n: the sample size; S: skewness; K: excess kurtosis (both 0 for the Normal distribution); $\chi^2_{0.95}(2)$: the 5% critical value
  - ν: degrees of freedom; $t_\nu^{-1}(\alpha)$: the α-quantile of the Student-t; $\sqrt{(\nu-2)/\nu}$ rescales it to variance 1
  - $L_{(i)}$: the i-th largest loss; k: the number of tail losses; p: the order of a moment; h: the number of days summed

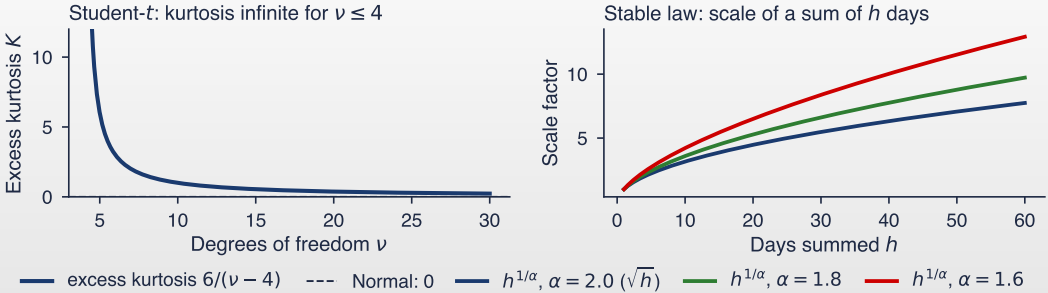
Prerequisites for Today (3/16): Returns and the Sharpe Ratio



- Left: $r = \ln(1+R)$ is below R and the gap grows with $|R|$; log returns add over time, $\sum_t r_t = \ln(P_n/P_0)$, so the total return is $e^{\sum r_t} - 1$
- Right: the lower end SR -1.96 SE of a 95% interval, with SE $= \sqrt{(1+\mathrm{SR}^2/2)/Y}$
  - the SE falls only as $1/\sqrt{Y}$: with SR $= 0.6$ about 13 years are needed before the interval excludes 0; with SR $= 0.3$, more than 40

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Prerequisites for Today (4/16): Kurtosis and Stable Scaling



- Left: a Student-$t(\nu)$ has excess kurtosis $K = 6/(\nu - 4)$ for $\nu > 4$; solving for ν gives $\nu = 4 + 6/K$: a large K means few degrees of freedom
- Right: under a stable law with index $\alpha \leq 2$, the sum of h days has the scale of one day times $h^{\mathbf{1}/\alpha}$
  - $\alpha = 2$ is the Normal case, $\sqrt{h}$; $\alpha < 2$: risk over h days grows faster than the square-root rule

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Prerequisites for Today (5/16): Efficiency, Volatility, Memory

Quantity	Formula
Ljung–Box	$Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(T-k) \sim \chi^2(m)$, $\chi^2_{0.95}(4) = 9.49$
Variance ratio	$\mathrm{VR}(q) = 1 + 2\sum_{k=1}^{q-1}(1-k/q)\hat{\rho}_k$; $Z = (\mathrm{VR}-1)/\sqrt{2(2q-1)(q-1)/(3qT)}$
Robust VR test	$Z^* = (\mathrm{VR}-1)/\mathrm{SE}_{rob}$; reject at 5% if $\lvert Z^*\rvert > 1.96$
EWMA	$\sigma^2_{t+1} = \lambda\sigma^2_t + (1-\lambda)r^2_t$, $\lambda = 0.94$
GARCH(1,1)	$\sigma^2_{t+1} = \omega + \alpha\varepsilon^2_t + \beta\sigma^2_t$, $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$, $h_{1/2} = \ln 0.5/\ln(\alpha+\beta)$
GARCH forecast	$E_t\sigma^2_{t+h} = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$
Hurst	$(R/S)_n \propto n^H$; risk over h days $\times h^H$

Prerequisites for Today (6/16): Notation for Efficiency, Volatility and Memory

- Dependence in time
  - $\hat{\rho}_k$: the autocorrelation at lag k; T: the number of returns; m: the number of lags tested; $\chi^2(m)$: the distribution of $Q(m)$ without autocorrelation
  - q: the horizon of the variance ratio, in days; VR $= 1$ under a random walk; SE_{rob}: a standard error robust to volatility clustering
- Volatility
  - σ_t^2: the variance forecast for day t; λ: the EWMA decay factor; ε_t: the shock (the demeaned return)
  - ω, α, β: GARCH parameters; $\bar{\sigma}^2$: the long-run variance; $h_{1/2}$: the half-life in days; E_t: the forecast made on day t for h days ahead
- Memory
  - $(R/S)_n$: the rescaled range on blocks of n days; H: the Hurst exponent, 0.5 without memory

Prerequisites for Today (7/16): Autocorrelation, Robust Variance Ratio and R/S

- **Sample autocorrelation** at lag k of T returns with mean $\bar{r}$:

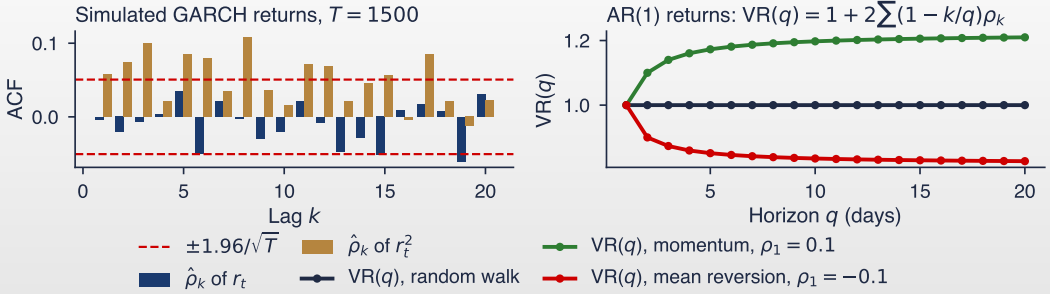
$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{T}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{T}(r_t - \bar{r})^2}, \qquad \hat{\rho}_k \approx N(0, 1/T) \text{ without autocorrelation}$$

- **Robust standard error** of $\widehat{\mathrm{VR}}(q)$ (Lo–MacKinlay), valid under volatility clustering:

$$\mathrm{SE}_{rob} = \sqrt{\hat{\theta}/T}, \qquad \hat{\theta} = \sum_{k=1}^{q-1}\left[\frac{2(q-k)}{q}\right]^2 \hat{\delta}_k, \qquad \hat{\delta}_k = \frac{T\sum_t (r_t - \bar{r})^2 (r_{t-k} - \bar{r})^2}{[\sum_t (r_t - \bar{r})^2]^2}$$

- **Rescaled range** on a block of n days with mean $\bar{x}$ and standard deviation S_n:
  - $Y_j = \sum_{i \le j}(x_i - \bar{x})$: the cumulated deviations; $(R/S)_n = (\max_j Y_j - \min_j Y_j)/S_n$, averaged over the blocks
  - H: the slope of $\log_{10}(R/S)_n$ on $\log_{10} n$; $H = 0.5$: no memory; $H > 0.5$: persistence

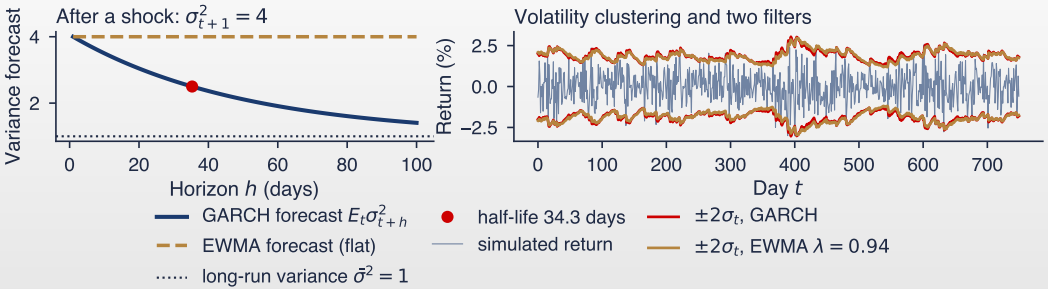
Prerequisites for Today (8/16): The ACF and the Variance Ratio



- Left: in simulated GARCH returns, $\hat{\rho}_k$ of r_t stays inside the band $\pm 1.96/\sqrt{T}$, while $\hat{\rho}_k$ of r_t^2 is positive at many lags: volatility clustering
- Right: $\mathrm{VR}(q) = 1$ for a random walk; above 1 with positive autocorrelation (momentum), below 1 with negative autocorrelation (mean reversion)

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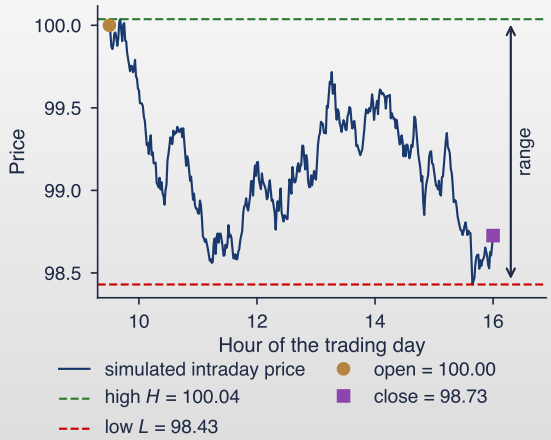
Prerequisites for Today (9/16): GARCH and EWMA Forecasts



- Left: after a shock, the GARCH forecast returns to $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$ at the rate $(\alpha+\beta)^{h-1}$, here with $\alpha+\beta = 0.98$; the EWMA forecast stays flat for every horizon
- Right: on simulated returns the two filters give almost the same daily σ_t; they differ in the forecasts for longer horizons

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Prerequisites for Today (10/16): The Daily Range

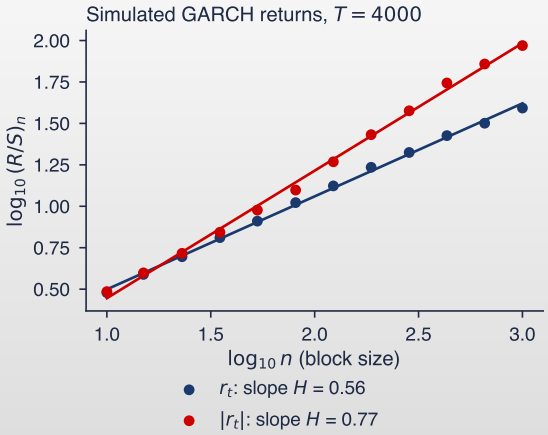


- A trading day: open, high H, low L, close
  - the close-to-close return uses two prices; the **range** $\ln H - \ln L$ uses the whole path of the day
- **Parkinson** variance of one day (log prices in %):

$$\hat{\sigma}_P^2 = \frac{(100 \ln H - 100 \ln L)^2}{4 \ln 2}$$

- Reading
  - $4 \ln 2$ makes $\hat{\sigma}_P^2$ unbiased for a Brownian motion without drift
  - annual volatility $\sqrt{252\, \hat{\sigma}_P^2}$; one day is a noisy estimate, so average over many days

Prerequisites for Today (11/16): Memory and the Hurst Exponent



- R/S analysis on simulated GARCH returns
  - each point: $\log_{10}$ of the average $(R/S)_n$ over the blocks of size n
  - the slope of the fitted line is the Hurst exponent H
- Reading
  - r_t: H close to 0.5, no memory in the returns (small blocks bias H slightly upwards)
  - $|r_t|$: H well above 0.5, long memory in the volatility
  - memory in $|r_t|$ does not make the sign of r_t predictable
  - scaling: risk over h days $\propto h^H$

Prerequisites for Today (12/16): Risk, Backtesting, Credit, Systemic Risk

Quantity	Formula
VaR, ES	$\mathrm{VaR}_\alpha = -q_\alpha > 0$; Normal: $-(\mu + \sigma z_\alpha)$, ES $-\mu + \sigma\varphi(z_\alpha)/\alpha$
Values	$z_{0.01} = -2.326$, $z_{0.025} = -1.960$, $\varphi(1.96) = 0.0584$, $t_5^{-1}(0.01) = -3.365$
Kupiec	$LR_{uc} = -2[(n-x)\ln(1-\alpha) + x\ln\alpha - (n-x)\ln(1-\hat{\pi}) - x\ln\hat{\pi}] \sim \chi^2(1)$, $\chi^2_{0.95}(1) = 3.84$
Logit, EL	$\mathrm{PD} = 1/(1+e^{-\eta})$, odds ratio e^{β_j}; EL = PD × LGD × EAD
AUC, Gini	$\mathrm{Gini} = 2\,\mathrm{AUC} - 1$
CoVaR	1% quantile regression $X_{sys} = a + bX_i$: $\mathrm{CoVaR} = -(a + b\,q^i_\alpha)$, $\Delta\mathrm{CoVaR} = b\,(q^i_{50} - q^i_\alpha)$

Prerequisites for Today (13/16): Notation for Risk, Credit and Systemic Risk

- Risk and backtesting
  - q_α: the α-quantile of the return; μ, σ: daily mean and standard deviation; z_α: the standard Normal quantile; φ: the standard Normal density
  - $t_{\mathbf{5}}^{-\mathbf{1}}(0.01)$: the 1% quantile of a Student-t with 5 degrees of freedom
  - x: the exceptions in n days; α: the target rate; $\hat{\pi} = x/n$: the observed rate; reject if $LR_{uc} > 3.84$
- Credit
  - $\eta = \beta_{\mathbf{0}} + \beta_{\mathbf{1}} x_{\mathbf{1}} + \ldots$: the linear score; PD: probability of default; e^{β_j}: the factor by which a one-unit rise in x_j multiplies the odds
  - EL: expected loss; LGD: loss given default; EAD: exposure at default; AUC: area under the ROC curve (0.5: no discrimination)
- Systemic risk
  - X_{sys}, X_i: the returns of the system and of bank i; a, b: intercept and slope; q_α^i, $q_{\mathbf{50}}^i$: the α-quantile and the median of the bank

Prerequisites for Today (14/16): Historical VaR and the Basel Traffic Light

- **Historical simulation** (HS) on the last m returns, sorted $r_{(1)} \le r_{(2)} \le \dots$:

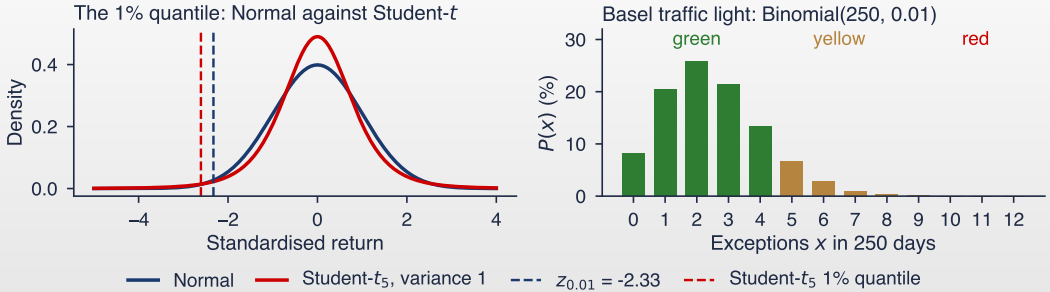
$$k = \lceil m\alpha \rceil, \qquad \mathrm{VaR}_\alpha = -r_{(k)}, \qquad \mathrm{ES}_\alpha = -\frac{1}{k}\sum_{i=1}^{k} r_{(i)}$$

- Notation
  - $\lceil \cdot \rceil$: round up; B2 uses $m = 500$ and moves the window forward one day at a time
  - **EWMA-Normal** VaR: $-(\mu + \sigma_{t+1} z_\alpha)$, with σ^2_{t+1} from EWMA; FHS (filtered HS): HS on the returns divided by σ_t
- **Basel traffic light**: for a correct VaR 1%, the exceptions in 250 days follow a Binomial distribution

$$P(X = x) = \binom{250}{x} 0.01^x \, 0.99^{250-x}, \qquad E[X] = 2.5$$

- Reading
  - green 0–4: the model is accepted; yellow 5–9: doubtful; red 10 or more: rejected (a correct model reaches it with probability of about 0.03%)
  - maximum in 250 consecutive days: the largest count over all windows of 250 days of the backtest

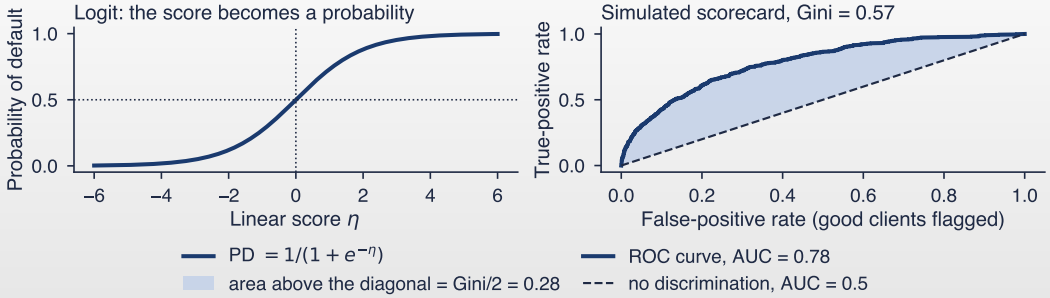
Prerequisites for Today (15/16): Heavy-Tailed Quantiles and the Traffic Light



- Left: with the same variance, the 1% quantile of a Student-t_5 is further out than $z_{0.01} = -2.33$: a Normal VaR 1% is too small for heavy tails
- Right: the Binomial(250; 0.01) probabilities: a correct model is green about 89% of the time and almost never red

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Prerequisites for Today (16/16): The Logit Curve and the ROC Curve



- Left: the logit turns the linear score η into a probability of default between 0 and 1; $\eta = 0$ gives 50%; a coefficient β_j multiplies the odds PD/(1 − PD) by e^{β_j}
- Right: the ROC curve plots the share of defaulters flagged against the share of good clients flagged, for every cut-off; AUC: the area under it; Gini = 2 AUC −1, twice the shaded area

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A1: Returns, Drawdown and the Sharpe Ratio [Solved]

Task

- A share closes at 50.0, 51.5, 49.8, 52.3, 47.9 and 50.6 RON on six consecutive days.
- 1. Compute the five simple and the five log returns, in %.
- 2. Compute the total return from the prices and from $\sum_t r_t$.
- 3. Compute the maximum drawdown and the gain needed to return to the peak.
- 4. A fund has a daily mean return of 0.04% and a daily standard deviation of 1.3%; with $r_f = 3\%$ a year, compute its annual mean, its annual volatility, its Sharpe ratio and the SE of the Sharpe ratio over 10 years.
- Report: fourteen numbers and one sentence.

Solution

- 1. R: 3.00, -3.30, 5.02, -8.41, 5.64; r: 2.96, -3.36, 4.90, -8.79, 5.48
- 2. $50.6/50 - 1 = 1.20\%$; $\sum r_t = 1.19\%$ and $e^{\mathbf{1.19/100}} - 1 = 1.20\%$
- 3. MDD $= 47.9/52.3 - 1 = -8.41\%$; gain $52.3/47.9 - 1 = 9.19\%$
- 4. $252 \times 0.04 = 10.08\%$; $\sqrt{252} \times 1.3 = 20.64\%$; SR $= (10.08 - 3)/20.64 = 0.343$; SE $= \sqrt{(1 + 0.343^{\mathbf{2}}/2)/10} = 0.33$
- Interpretation: the Sharpe ratio is about one SE from zero: ten years of data cannot show that this fund beats the risk-free rate.

Back Exercise list

A2: Normality, Tails and Stable Scaling [Proposed]

Solution discussed in class

Task

- $n = 2000$ daily returns: skewness $S = -0.8$, excess kurtosis $K = 9$. The six largest losses, in %: 11.2, 8.4, 7.7, 6.0, 5.5, 4.9. Model: Seminar 2, A7 and Seminar 5, A1.
- 1. Compute JB, its skewness and kurtosis parts, and decide at 5%.
- 2. Compute ν of a Student-t distribution with the same excess kurtosis.
- 3. Compute the Hill estimate $\hat{\alpha}$ with $k = 5$ and its SE.
- 4. A stable law with $\alpha = 1.6$ describes daily returns: compute the 20-day scale factor and compare it with $\sqrt{20}$.
- Report: seven numbers, one decision and one sentence.

Back Exercise list

A3: Autocorrelation and the Variance Ratio [Solved]

Task

- $T = 1500$ daily returns; $\hat{\rho}_1 = -0.05$, $\hat{\rho}_2 = 0.03$, $\hat{\rho}_3 = 0.02$, $\hat{\rho}_4 = -0.01$; the robust SE of $\widehat{\mathrm{VR}}(5)$ is 0.055.
- 1. Compute the Ljung–Box statistic $Q(4)$ and decide at 5%.
- 2. Compute $\mathrm{VR}(5)$.
- 3. Compute $Z(5)$ and $Z^*(5)$ and decide at 5%.
- Report: four numbers, two decisions and one sentence.

Solution

- 1. $\sum \hat{\rho}_k^2 = 0.0039$; $Q(4) = 5.86 < 9.49$ ($p = 0.210$): not rejected
- 2. $1 + 2(-0.040 + 0.018 + 0.008 - 0.002) = 1 + 2 \times (-0.016) = 0.968$
- 3. SE $= \sqrt{72/22\,500} = 0.0566$: $Z = -0.57$; $Z^* = (0.968 - 1)/0.055 = -0.58$; both $|Z| < 1.96$: not rejected
- Interpretation: no evidence against the random walk at one week; weak-form efficiency is not rejected.

Back Exercise list

A4: Volatility Forecasts [Proposed]

Solution discussed in class

Task

- GARCH(1,1), returns in %: $\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$; today $\sigma_t^2 = 2.5$ and $\varepsilon_t = 1.0$. One day of an index: high 105.2, low 101.6. Model: Seminar 9, A1 and A3; Seminar 8, A3 and A5.
- 1. Compute σ_{t+1}^2 and the EWMA forecast with $\lambda = 0.94$.
- 2. Compute the persistence, the half-life, the long-run variance and the long-run annual volatility.
- 3. Compute $E_t[\sigma_{t+5}^2]$.
- 4. Compute the Parkinson variance of the day and the annualised Parkinson volatility.
- Report: eight numbers and one sentence.

Back Exercise list

A5: VaR, ES and Backtesting [Proposed]

Solution discussed in class

Task

- A position of 2 000 000 RON; daily returns with mean 0.05% and standard deviation 1.2%. Model: Seminar 10, A1, A3 and A5.
- 1. Compute the Normal VaR 1% and ES 2.5%, in % and in RON.
- 2. Compute VaR 1% if the standardised returns are Student-t with $\nu = 5$.
- 3. A model has 17 exceptions of VaR 1% in 1000 days: compute LR_{uc} and decide at 5%.
- 4. Give the Basel zone of 11 exceptions in 250 days.
- Report: seven numbers, two decisions and one sentence.

Back Exercise list

A6: Credit Scoring and CoVaR [Proposed]

Solution discussed in class

Task

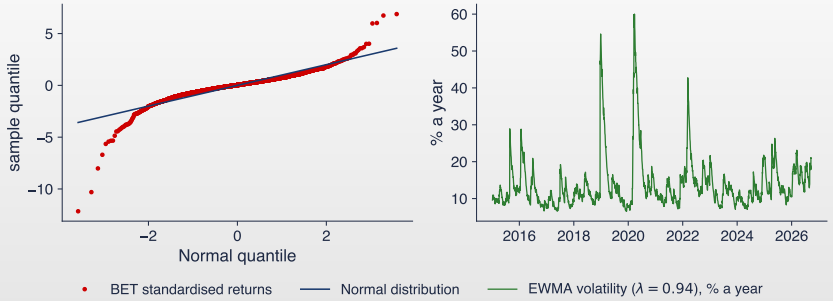
- Logit: $\eta = -1.8 + 0.9x_{\mathbf{1}} - 0.6x_{\mathbf{2}}$, applicant with $x_{\mathbf{1}} = 1.2$, $x_{\mathbf{2}} = 2.5$; LGD $= 40\%$, EAD $= 150\,000$ RON. A 1% quantile regression of the system return on a bank: $a = -1.8$, $b = 0.45$; bank: $q_{\mathbf{0.01}} = -6.0\%$, median 0.05%. Model: Seminar 12, A1 and A5; Seminar 15, A1.
- 1. Compute the PD of the applicant and the odds ratio of $x_{\mathbf{1}}$.
- 2. Compute the expected loss of the loan.
- 3. A scorecard has Gini $= 0.56$: compute its AUC.
- 4. Compute CoVaR 1% at the bank's VaR and at its median, and ΔCoVaR.
- Report: seven numbers and one sentence.

Back Exercise list

B1: A Full Check-Up of the BET [Solved]

- **Question**: what do Chapters 1–11 say about the BET since 2015?
- **Inputs**: BET closes since 2 January 2015, daily log returns in %
- **Tasks**
  1. Compute the annual mean, the volatility, the Sharpe ratio with its SE and the maximum drawdown.
  2. Compute the skewness, the excess kurtosis, JB and the Hill tail index of the losses with $k = 2.5\%$ of n.
  3. Compute Ljung–Box $Q(10)$ on r_t and on r_t^2, and fit a GARCH(1,1)-t.
  4. Compute VaR 1% and ES 2.5% by historical simulation and with the Normal distribution.
  5. Interpretation: which three facts would you write in an exam answer about the BET?
- **Report**: a table of about fifteen numbers, the chart and three sentences
- Notebook: section B1

B1: Solution (1/2) [Solved]



- $n = 2\,925$, 6 January 2015 – 18 September 2026; critical value $\chi^2_{0.95}(10) = 18.31$; EWMA volatility peak 60.0% on 25 March 2020

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B1: Solution (2/2) [Solved]

Performance		**Distribution, dependence**	
mean, % a year	13.2	skewness; excess kurtosis	-1.41; 17.8
volatility, % a year	15.5	JB; Hill $\hat{\alpha}$ (SE)	39 794; 2.19 (0.26)
Sharpe (SE)	0.93 (0.35)	$Q(10)$: r_t; r_t^2	54.7; 701
MDD	-31.1%	GARCH-t: $\alpha+\beta$; $h_{1/2}$	0.925; 8.8
VaR 1%: HS; Normal	2.72; 2.23	ES 2.5%: HS; Normal	3.21; 2.24

- Interpretation: (1) heavy tails: the Normal VaR 1% is 18% too low; (2) returns are weakly autocorrelated, squared returns strongly: volatility clustering; (3) the Sharpe ratio is less than three SE from zero

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B2: Backtesting VaR 1% for the S&P 500 and Bitcoin [Proposed]

- **Question**: does a heavy-tailed method beat a Normal method with fast volatility? Model: B1 and Seminar 10, B3.
- **Inputs**: S&P 500 and Bitcoin, all data; forecasts for every day since 1 January 2017
- **Tasks**
  1. Compute the one-day VaR 1% by historical simulation on the last 500 returns.
  2. Compute the one-day EWMA-Normal VaR 1% with $\lambda = 0.94$.
  3. Count the exceptions of each method and run the Kupiec test.
  4. Find the largest number of exceptions in any 250 consecutive days.
  5. Interpretation: which method would a supervisor accept for each series?
- **Report**: one table and two sentences
- Notebook: section B2

B3: Efficiency and Memory in Two Windows [Proposed]

- **Question**: did the BET and the S&P 500 become more efficient after 2015? Model: A3; Seminar 7, B1 and Seminar 11, B1.
- **Inputs**: BET and S&P 500, two windows: 2000–2014 and 2015–2026
- **Tasks**
  1. Compute VR(5) and the robust $Z^*(5)$ in each window.
  2. Compute the Hurst exponent (R/S) of r_t and of $|r_t|$ in each window.
  3. Interpretation: did the BET become more efficient after 2015?
- **Report**: one table of sixteen numbers and two sentences
- Notebook: section B3

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to summarise the course for the exam, with the course data. The answer:
- (a) The VaR 99% of the BET since 2015 is −2.72%.
- (b) The BET returns are close to Normal: the skewness is only −1.41.
- (c) Bitcoin has a daily standard deviation of 3.49%, so its annual volatility is 55.4%.
- (d) For the S&P 500, alpha + beta = 0.9928, so a volatility shock halves in 1/(1 - 0.9928) = 139 days.
- (e) The Hurst exponent of |r| is 0.84 for the S&P 500, so its returns are predictable and the market is inefficient.
- (f) For Normal returns, ES 2.5% and VaR 1% are almost the same number.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

What You Should Take from Today

- Every exam problem has three parts: the formula with the numbers, the result with its unit and sign, the interpretation
- Tests: state H_0, the statistic, the critical value, the decision; prefer the robust version
- Risk: VaR 1% is a positive loss; the Normal distribution understates it on real data
- Volatility is predictable, the sign of returns almost not
- An AI answer is a draft: check the level and the sign of VaR, the annualisation factor and every formula
- Next
  - Lecture 16: the course map, two summary slides per chapter, the toolbox, eight solved exam-type problems, the project
  - Try the [Proposed] tasks on paper first, then check them in the notebook; B3 can grow into a team project
  - Reading: Franke, Härdle & Hafner (2019); exercises in Borak, Härdle & López-Cabrera (2013)

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